Sensor Data Quality Investigation

Determine whether a sensor data set is accurate, precise, biased, or affected by noise.

Advanced	2 class periods	Engineering Challenge

Engineering Context

A rover environmental sensor is producing inconsistent readings. Before replacing hardware, the test team must characterize the data and identify likely error sources.

What You Will Practice

- Distinguish accuracy and precision
- Calculate mean, range, and standard deviation
- Detect bias and outliers
- Recommend a calibration or filtering method

What You Need

- Sensor or instructor-provided data set
- Reference standard
- Spreadsheet
- Graphing tools
- Engineering notebook

Student Instructions

1. Collect or receive at least 20 readings under a constant condition.
2. Record a trusted reference value.
3. Calculate mean, range, standard deviation, and error of the mean from the reference.
4. Create a time-series graph and a histogram or dot plot.
5. Identify any outliers using an instructor-approved rule.
6. Repeat measurements after one controlled change such as repositioning, shielding, or recalibration.
7. Compare the two data sets.
8. Recommend keep, calibrate, filter, reposition, or replace.

Engineering Requirements

- At least 20 readings per primary condition
- Reference value clearly identified
- Include numerical measures of center and spread
- Do not remove outliers without documenting the rule
- Recommendation must be supported by evidence

Constraints & Safety

- Do not expose sensors to unsafe conditions
- Do not fabricate readings
- Keep environmental variables controlled unless intentionally tested

What You Submit

- Raw data
- Calculations
- Two graphs
- Outlier decision
- Before/after comparison
- Sensor recommendation

Reflection Questions

1. Was the main problem bias or variation?
2. What evidence supported removing or retaining an outlier?
3. Which controlled change improved data quality most?
4. What filter could reduce noise without hiding real changes?

Engineering Tip Plot data in time order before summarizing it; drift may disappear in a simple average.	**Common Mistake** Removing all unusual values because they look inconvenient can hide a real fault.
Stretch Goal Implement and compare a moving-average filter with two different window sizes.	**Career Connection** Instrumentation and test engineers validate sensors used for navigation, structures, propulsion, and environmental monitoring.